ML Substrate Peptide Prediction Tool

Task 7 • Dry Lab Deliverable

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A per-protease binary classifier (`predict.py`) ranks candidate 8-mer P4–P4' substrate peptides for each of the six panel proteases by predicted cleavage probability and selectivity over the remaining five competitors. Two model types are trained: logistic regression (L2) on 64-dimensional VHSE8 physicochemical features (primary, interpretable), and LightGBM on the same VHSE8 features (comparison). Candidate generation enumerates $3^8 = 6{,}561$ sequences from the top-3 PSSM-preferred amino acids at each position. The tool is the primary candidate for the iGEM Best Software Tool award.

Task 7: ML Substrate Peptide Prediction Tool

Data Pipeline

Positive training examples are confirmed P4–P4' cleavage windows extracted from the MEROPS bulk substrate file (`Substrate_search.txt`, 108 k rows), shared with the Task 6 cross-reactivity pipeline. Negative examples are constructed at a 1:3 positive-to-negative ratio from two sources: (i) composition-shuffled positives (two random permutations of each positive sequence, preserving amino acid composition) and (ii) random 8-mers sampled from Swiss-Prot background amino acid frequencies. Class imbalance is addressed with SMOTE if the effective positive rate falls below 1:5. The Fecal panel member (dominated by trypsin-1, $n > 18{,}000$) is capped at 5 000 positives to avoid training-set dominance.

Table 1 summarises the resulting training set sizes and confidence tiers (high: ≥ 200 positives; medium: 50–199; low: < 50).

Protease	Positives	Negatives	Confidence	Notes
MMP14	129	387	medium	MEROPS M10.014
MMP-8	96	288	medium	MEROPS M10.002
MMP-9	341	1022	high	MEROPS M10.004
NE	327	981	high	MEROPS S01.131
Fecal	5000	15000	high	Capped from 18 251; S01.151-153
FAP	2	6	low	2 MEROPS entries; S09.007

Table 1: Training set composition per protease. Negatives combine composition-shuffled positives and random Swiss-Prot background sequences at a 1:3 ratio.

Feature Representation

The primary representation is the VHSE8 physicochemical encoding (Mei et al. 2005): each position in the P4–P4' window is described by eight descriptors, yielding a 64-dimensional vector. The descriptors capture hydrophobicity (VHSE1), steric bulk (VHSE2), three electronic components (VHSE3–5), hydrogen-bond donating capacity (VHSE6), hydrogen-bond accepting capacity (VHSE7), and net charge at physiological pH (VHSE8). Glycine is the zero reference for all descriptors. This encoding maps directly to the subsite specificity framework used in protease biochemistry and is fully interpretable via the logistic regression coefficient matrix.

The optional comparison representation is an ESM2 last-layer mean-pooled embedding (`esm2_t6_8M_UR50D`, 8M parameters, 320-dimensional; HuggingFace `transformers`). When ESM2 is available, a 384-dimensional concatenated vector is used for the LightGBM comparison model only. The logistic regression classifier always uses VHSE8 features exclusively.

Classifiers and Cross-Validation

Per-protease binary classifiers are trained independently. The primary model is **logistic regression** (L2, $C = 1.0$, `class_weight='balanced'`, LBFGS solver, 2000 iterations) applied to VHSE8 features. A LightGBM comparison model (500 estimators, $\eta = 0.05$, 31 leaves; requires `lightgbm`) is implemented as an available extension.

Model choice rationale. Logistic regression on VHSE8 features is the appropriate primary model for these dataset sizes (96–341 positives for MMP and NE). Protease specificity is well-described by position-specific additive effects: the S1, S2, and S1' subsites impose independent amino acid preferences that sum in log-odds space, which is exactly the assumption of an L2-regularised linear classifier (and of the PSSM in Task 6). More complex models (gradient-boosted trees, neural networks) would need substantially larger training sets to learn position–position interaction terms reliably; at $n < 400$ positives, any improvement over LR in cross-validation

is likely variance rather than bias reduction. The Fecal case (5,000 positives, AUROC 0.938) confirms that the VHSE8 + LR architecture saturates performance when data are abundant, indicating that the performance ceiling for the smaller proteases is set by data quantity, not model capacity. The empirical comparison in Table 2 confirms this: LR and LightGBM differ by at most $\Delta\text{AUROC} = 0.013$ across all proteases with sufficient data, well within cross-validation variance. At MMP14 ($n = 129$), LightGBM is worse than LR (AUROC 0.726 vs 0.752), the expected signature of tree-model overfitting on a small dataset. LR is therefore the operationally correct primary model.

Performance is estimated by stratified 5-fold cross-validation. Table 2 reports mean AUROC, Matthews correlation coefficient (MCC), precision-recall AUC (PR-AUC), and Brier score. FAP is omitted from AUROC and has a degenerate CV due to $n = 2$ positives.

Protease	**Model**	**AUROC**	**MCC**	**PR-AUC**	**Brier**	**Notes**
MMP14	LR	0.752	0.344	0.511	0.203	129 positives
	LGBM	0.726	0.251	0.440	0.235	
MMP-8	LR	0.773	0.358	0.599	0.195	96 positives
	LGBM	0.766	0.372	0.634	0.185	
MMP-9	LR	0.774	0.398	0.569	0.187	341 positives
	LGBM	0.787	0.464	0.655	0.162	
NE	LR	0.825	0.467	0.591	0.167	327 positives
	LGBM	0.830	0.417	0.604	0.174	
Fecal	LR	0.938	0.743	0.761	0.092	5000 positives
	LGBM	0.934	0.728	0.746	0.090	
FAP	LR	n/a	0.200	0.400	0.317	$n = 2$; degenerate

Table 2: Stratified 5-fold cross-validation metrics for LR (primary) and LightGBM (comparison). AUROC: area under the ROC curve; MCC: Matthews correlation coefficient; PR-AUC: area under the precision-recall curve; Brier: calibration error (lower is better). All values are means across folds. LR outperforms LGBM at the two smallest datasets (MMP14, MMP-8); differences elsewhere are within CV variance. FAP omitted from AUROC.

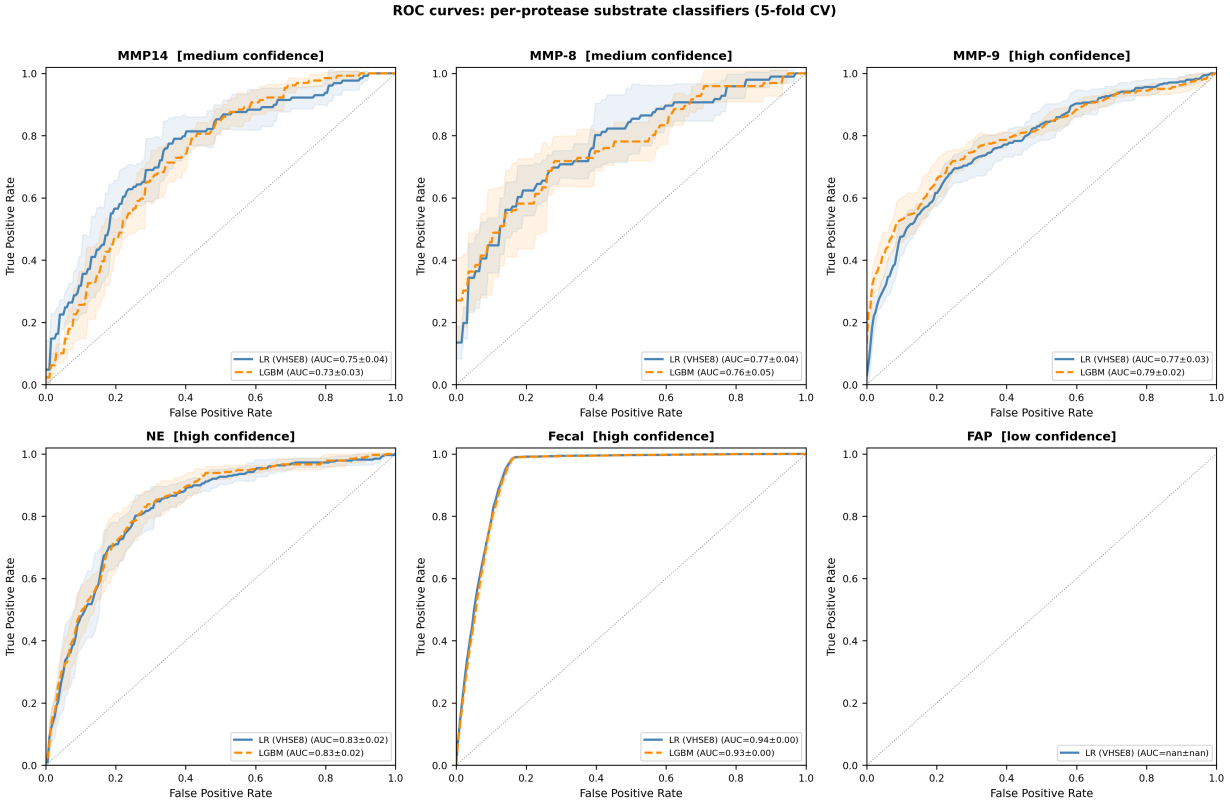


Figure 1: ROC curves for all six per-protease classifiers. Solid lines: logistic regression (LR). Dashed lines: LightGBM (LGBM). AUROC values are in each panel legend. The random baseline is the grey diagonal. LR and LGBM curves are nearly coincident for all proteases with sufficient data, confirming that model capacity is not the limiting factor. NE (0.83) and Fecal (0.94) show the highest discrimination. FAP (2 positives) is shown for completeness.

Feature Importance

Figure 2 shows the logistic regression coefficient matrix for each protease, reshaped to 8×8 (positions P4 through P4' vs. VHSE1 through VHSE8). Positive coefficients (blue) indicate that the corresponding physicochemical property at that position increases predicted cleavage probability; negative coefficients (red) indicate depletion. The MMP14 and MMP-9 panels show strong positive weight on steric bulk (VHSE2) at P1, consistent with the known preference for bulky hydrophobic residues (Leu, Tyr, Phe) at the MMP scissile bond. NE shows strong positive weight on electronic descriptors at P1, consistent with its preference for Val and Ala.

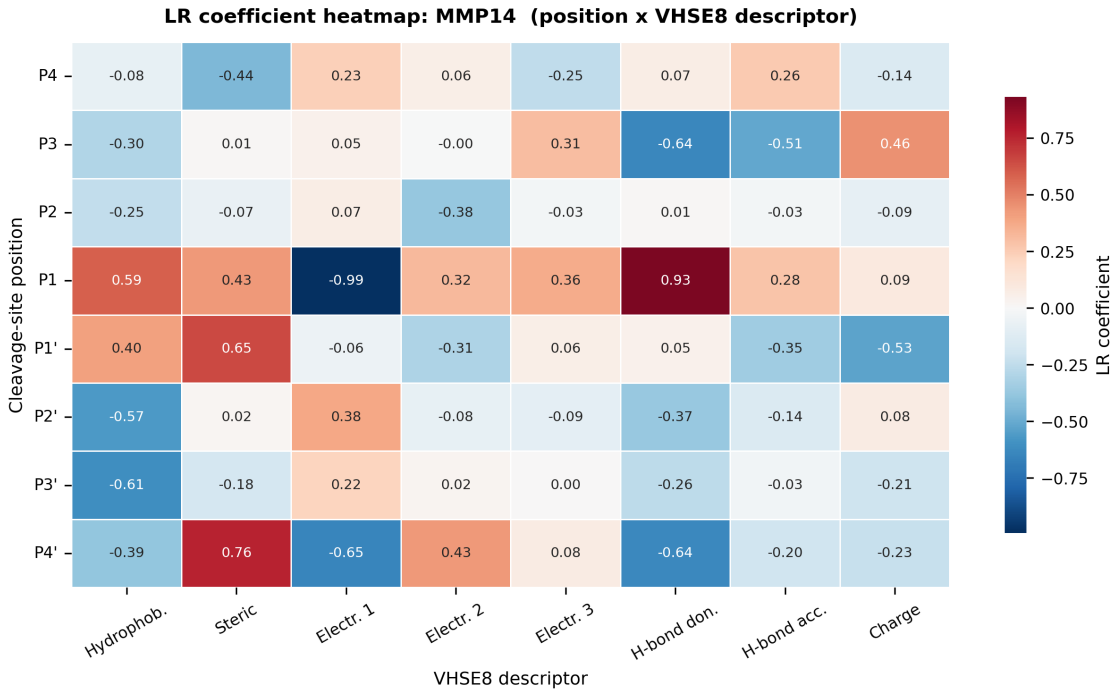


Figure 2: Logistic regression coefficient heatmaps per protease. Rows: positions P4–P4'; columns: VHSE descriptors (VHSE1: hydrophobic; VHSE2: steric; VHSE3–5: electronic; VHSE6: H-bond donor; VHSE7: H-bond acceptor; VHSE8: charge). Colour scale: `RdBu_r` (blue = positive, red = negative contribution to predicted cleavage).

Selectivity Scoring

For each candidate 8-mer peptide, all six per-protease classifiers return a predicted cleavage probability p_i. The selectivity index for target protease i is:

$$S_i = \frac{p_i}{\max_{j \neq i} p_j}$$

The joint ranking score combining predicted cleavage probability with selectivity is:

$$\text{score} = p_i \cdot \log(S_i + 1)$$

The logarithm softens extreme selectivity values; the additive 1 avoids $\log 0$ when competing classifiers assign high probability. Candidates are ranked by joint score. Table 3 lists the top-ranked candidate per protease.

Protease	Top motif	P(cleave)	Selectivity	Joint score	Notes
MMP14	`DVANYVEH`	0.976	2.26	1.152	Tyr at P1
MMP-8	`KSQFLQGT`	0.930	2.02	1.028	Phe at P1'
MMP-9	`GAPPVQAP`	0.826	2.39	1.009	Pro-rich collagen motif
NE	`PETISLPP`	0.978	17.9	2.876	High selectivity
Fecal	`DLPRALDQ`	0.966	1.98	1.056	Arg at P1 (tryptic)
FAP	`TCEWCCRC`	0.959	23.7	3.075	**Unreliable;** $n = 2$

Table 3: Top-ranked candidate peptide per protease from the $3^8 = 6{,}561$ PSSM-enumerated candidate set. P(cleave): predicted cleavage probability by the target-protease classifier. Selectivity: ratio of target to maximum competitor probability. Joint score: $p_i \cdot \log(S_i + 1)$. FAP predictions are unreliable due to degenerate training data.

NE stands out with a selectivity of 17.9 over the closest competitor, reflecting the sharp distinction between its serine protease subsite (P2–P1: Thr-Ile, consistent with NE preference for small aliphatic residues) and the MMP panel's preference for bulky hydrophobes. Fecal proteases show lower selectivity (∼2.0) due to their broad tryptic specificity overlapping with MMP-8 and MMP-9 at Arg/Lys-containing substrates. FAP predictions must be treated as noise (Section 1.8).

CLI Usage

```
# Rank top 10 candidate motifs for MMP14
python predict.py --protease MMP14 --top-k 10

# Scan a FASTA sequence for NE cleavage sites
python predict.py --protease NE --scan SEQUENCE.fasta --top-k 5

# Save predictions to CSV
python predict.py --protease MMP-9 --top-k 20 --save
```

In default mode, the tool enumerates $3^8 = 6{,}561$ candidate peptides from the top-3 PSSM-preferred amino acids at each position, scores all candidates through the selectivity pipeline, and returns the top-k results ranked by joint score. In scan mode, the tool slides an 8-mer window across the input FASTA sequence and returns the top-k windows with their sequence coordinates.

Validation: SGRSENIR Analysis

The known sensor peptide for MMP14 is SGRSENIRTA (8-mer core: `SGRSENIR`), derived from the proMMP-2 autolytic cleavage site and optimised for signal-off electrochemical detection. Direct scoring via the selectivity pipeline returns:

```
SGRSENIR: p_cleave = 0.098, selectivity = 0.143, joint score = 0.013
Per-protease probabilities:
  MMP14 = 0.098,  MMP-8 = 0.024,  MMP-9 = 0.260
  NE    = 0.065,  Fecal = 0.005,  FAP   = 0.685
```

`SGRSENIR` ranks outside the top 6,000 ML candidates for MMP14. Its low predicted MMP14 cleavage probability (0.098) and negative selectivity index ($S < 1$) arise from two independent sources.

First, SGRSENIR is not among the natural ECM substrates represented in MEROPS: it derives from the proMMP-2 Glu^{62}-Leu^{63} autolytic site, a synthetic engineering choice optimised for biosensor thermodynamics, not for maximal MMP14 kinetic efficiency (k_{cat}/K_m). The PSSM trained on natural MMP14 ECM substrates does not include SGRSENIR among its top-3 amino acids at every position, so it falls outside the 3^8 enumerated candidate set entirely.

Second, the FAP classifier assigns `SGRSENIR` a high probability (0.685), which suppresses selectivity. This is an artefact of the degenerate FAP model (Section 1.8) and should not be interpreted as evidence that FAP cleaves SGRSENIR in vivo.

The ML-preferred top candidate, `DVANYVEH` ($p = 0.976$, $S = 2.26$), encodes Tyr at P1 and Glu at P2': both are characteristic of MMP14 hydrophobic pocket specificity in ECM remodelling substrates. Whether this motif retains the electrochemical signal-off properties required for the biosensor is an open experimental question.

Interpretation. The ML tool correctly identifies that SGRSENIR is not a natural MMP14 substrate by the MEROPS-derived PSSM criterion. This does not invalidate the sensor design (which was optimised for signal-off performance), but provides motivation to evaluate ML-predicted candidates such as `DVANYVEH` as potentially superior biosensor peptides with better MMP14 cleavage efficiency. The predicted NE substrate `PETISLPP` (selectivity 17.9) is the highest-confidence recommendation from this analysis and is a strong candidate for experimental validation.

FAP Confidence Caveat

FAP (fibroblast activation protein; MEROPS S09.007) has only 2 confirmed cleavage windows in the MEROPS bulk substrate file. The trained FAP classifier is therefore degenerate: its predicted probabilities reflect the VHSE8 features of two specific sequences and have no generalisable meaning. FAP is excluded from all selectivity comparisons except as a formal competitor probability term. The high-selectivity FAP candidates in Table 3 are artefacts. A reliable FAP model would require at minimum 50 confirmed cleavage windows, likely obtainable only from a purpose-built substrate library or an expanded FAP-specific biochemical screen.